

Typesetting lecture notes with `multinotes.sty`

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Contents

1	Options	2
2	Inline boxes	2
2.1	<code>\blank</code>	2
2.2	<code>\fillbox</code>	3
2.3	<code>\ans</code>	3
2.4	<code>\prf</code>	3
3	Display boxes	3
3.1	The <code>blankbox</code> environment	3
3.2	The <code>answer</code> environment	4
3.3	The <code>proof</code> environment	5
4	Sketchboxes	6
5	Videos	7
5.1	<code>\includevideo</code>	7
5.2	The <code>video</code> environment	7
6	Images	7
6.1	<code>\includegraphics</code>	7
6.2	<code>\includetwographics</code>	8
6.3	<code>TikZ</code>	8

1 Options

The following package options are available.

Option	Effect
<code>blanks</code>	<code>\blank</code> macros and <code>blankbox</code> environments replaced by whitespace
<code>noproofs</code>	<code>\prf</code> macros and <code>proof</code> environments excluded
<code>blankproofs</code>	<code>\prf</code> macros and <code>proof</code> environments replaced by whitespace
<code>noanswers</code>	<code>\ans</code> macros and <code>answer</code> environments excluded
<code>blankanswers</code>	<code>\ans</code> macros and <code>answer</code> environments replaced by whitespace
<code>nosketchboxes</code>	<code>sketchbox</code> environments excluded

The following style commands are available.

Command	
<code>\textcolour{}</code>	set text colour
<code>\backgroundcolour{}</code>	set background colour
<code>\showcolour{}</code>	set text colour of blank boxes
<code>\proofcolour{}</code>	set text colour of proofs
<code>\answercolour{}</code>	set text colour of answer boxes
<code>\framedblankboxes</code>	framed blankboxes
<code>\framedproofs</code>	framed proofs
<code>\framedanswers</code>	framed answers
<code>\setboxrule{}</code>	set frame thickness
<code>\setstretchfactor{}</code>	to accomodate handwriting in blank boxes
<code>\setimagestretchfactor{}</code>	to accomodate hand-drawing in blank boxes

2 Inline boxes

The following macros replace their content by whitespace in response to different options. These can be used for partial handouts to accompany lectures or videos, or for missing-word exercises etc.

- `\blank{...}` is emptied by the `blanks` option.
- `\fillbox{...}` is emptied by `blankanswers` and `noanswers`.
- `\ans{...}` is emptied by `blankanswers` and removed by `noanswers`.
- `\prf{...}` is emptied by `blankproofs` and removed by `noproofs`.

2.1 `\blank`

Inline text replaced by whitespace by the `blanks` option.

Here is some `\blank{text}` to be hidden.

Here is some text to be hidden.

2.2 `\fillbox`

Similar to `\blank` but controlled by `blankanswers`.

```
Roses are \fillbox{red}, violets are \fillbox{blue}.
```

Roses are red, violets are blue.

2.3 `\ans`

Similar to `\fillbox` but also controlled by the `noanswers` option.

```
What is $7\times 8$? \ans{54}
```

What is 7×8 ? 54

2.4 `\prf`

Similar to `\ans` but controlled by `blankproofs` and `noproofs`.

```
Thm:  $\sin^2\theta + \cos^2\theta = 1$ . \prf{This follows by Pythagoras' theorem.}
```

Thm: $\sin^2\theta + \cos^2\theta = 1$. This follows by Pythagoras' theorem.

3 Display boxes

- `\begin{blankbox}...\end{blankbox}` is controlled by the `blank` option.
- `\begin{proof}...\end{proof}` is controlled by `blankproofs` and `noproofs`.
- `\begin{answer}...\end{answer}` is controlled by `blankanswers` and `noanswers`.

3.1 The blankbox environment

3.1.1 A simple blankbox

```
\begin{blankbox}  
\lipsum[1]  
\end{blankbox}
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

3.1.2 A blankbox with an optional title

```

\begin{blankbox}[\sl Some ideas]
\lipsum[2]
\end{blankbox}

```

Some ideas Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

3.1.3 A blankbox inside a theorem-like environment

Example 1. If $X \sim \text{Poisson}(\lambda)$, compute $P(X \leq 2)$.

Solution

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{so} \quad P(X \leq 2) = \left(1 + \lambda + \frac{\lambda^2}{2}\right) e^{-\lambda}.$$

```

\begin{example}
If  $X \sim \text{Poisson}(\lambda)$ , compute  $P(X \leq 2)$ .
\begin{blankbox}[\sl Solution]
\[
P(X=k) = \displaystyle \frac{\lambda^k e^{-\lambda}}{k!}
\quad \text{so} \quad
P(X \leq 2) = \left(1 + \lambda + \frac{\lambda^2}{2}\right) e^{-\lambda}.
\]
\end{blankbox}
\end{example}

```

3.2 The answer environment

- `blankanswers` replaces the contents by blank space.

- `noanswers` removes it completely.

The `noanswers` option is implemented using the `comment` package. For it to work properly, the opening and closing commands `\begin{answers}` and `\end{answers}` must each appear on a line of its own, with no additional whitespace. This also applies to the `proof` environment with the `noproofs` option.

Here is an answer (might be removed).

Answer Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

This is produced by the following code.

```
\begin{answer}
\lipsum[3]
\end{answer}
```

Here is an answer with a custom title (might be removed).

Answer to exercise 3 Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula. $\square$

This is produced by the following code.

```
\begin{answer}[\sl Answer to exercise 3]
\lipsum[4]
\qed
\end{answer}
```

3.3 The proof environment

- `blankproofs` replaces the contents by blank space.
- `noproofs` removes the proof completely.

Here is a proof (might be removed).

Proof Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi

in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

This is produced by the following code.

```
\begin{proof}  
\lipsum[5]  
\end{proof}
```

Here is a proof with a custom title (might be removed).

Proof of main theorem Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

This is produced by the following code.

```
\begin{proof}[\sl Proof of main theorem]  
\lipsum[6]  
\end{proof}
```

4 Sketchboxes

The `sketchbox` command produces an empty box for drawing sketches during live sessions. The mandatory argument determines the height of the box as a proportion of `\textheight`. The optional argument provides a title for the box.

Sketch (Draw your sketch here)

This is produced by the following code:

```
\sketchbox[Draw your sketch here]{0.15}
```

5 Videos

The `\includevideo` macro and `video` environment are incorporated in anticipation of an effective LaTeX to HTML conversion method. For PDF output, video content is rendered as a simple hyperlink.

5.1 `\includevideo`

The `\includevideo` command mirrors `\includegraphics`.

<https://youtu.be/wqGUwLOP6no>

This is produced by the following code:

```
\includevideo[scale=0.5]{https://youtu.be/wqGUwLOP6no}
```

5.2 The video environment

A new float called `video` mimics the `figure` and `table` environments. These can be framed and captioned. A corresponding `\listofvideos` (suffix `.lov`) can be included.

<https://youtu.be/wqGUwLOP6no>

Video 5.1: Optimistic voices

This is produced by the following code:

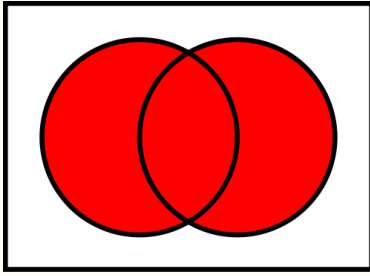
```
\begin{video}  
\centering  
\includevideo[scale=0.5]{https://youtu.be/wqGUwLOP6no}  
\caption{Optimistic voices\label{vid:optimistic}}  
\end{video}
```

6 Images

6.1 `\includegraphics`

`\includegraphics{img}` replaces `img` by whitespace whenever the `blanks` option is set.

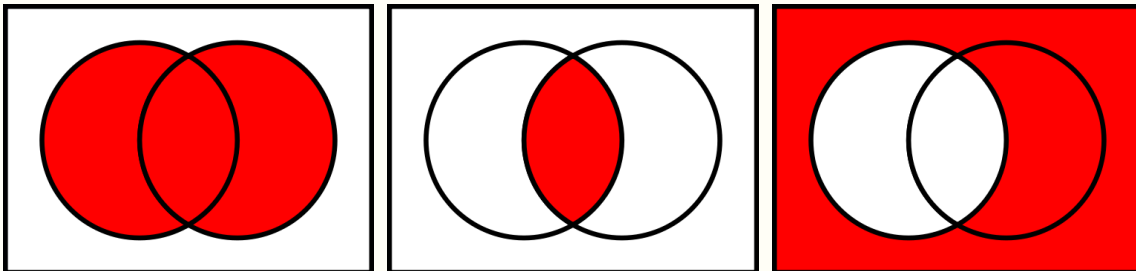
```
\begin{blankbox}  
\resizebox{0.3\textwidth}{!}{\includegraphics{AcupB}}  
\end{blankbox}
```



6.2 `\includetwographics`

`\includetwographics{img1}{img2}` shows `img2` whenever `blanks`, `blankproofs` or `blankanswers` is set, otherwise it shows `img1`. This can be used to print partial images in handout notes and exercise sheets.

```
\begin{blankbox}
\resizebox{0.3\textwidth}{!}{\includetwographics[scale=1]{AcupB}{emptyset}}
\resizebox{0.3\textwidth}{!}{\includetwographics[scale=1]{AcapB}{emptyset}}
\resizebox{0.3\textwidth}{!}{\includetwographics[scale=1]{Acomp}{emptyset}}
\end{blankbox}
```



6.3 TikZ

Here is a tikz picture in a blankbox. The `\blanktikz` command should make it disappear when the `blanks` option is set.

```
\begin{blankbox}
\blanktikz
\resizebox{0.5\textwidth}{!}{\tikz\pic{rwalk};}
\end{blankbox}
```

